

Temporal Fractal Coupling of Volatility and Trading
Volume:
A Robust Within-Firm Pattern, a Time-Period Stress
Signal, and No Firm-Specific Forecast Power in
S&P 500 Stocks, 2015–2026

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Abstract

Do the fractal scaling properties of price volatility and trading volume co-evolve, and does their co-movement carry predictive information about future market conditions? I apply detrended fluctuation analysis to 488 current S&P 500 constituents over 2015–2026, computing rolling Hurst exponents for absolute returns ($H_{|r|,t}^{(i)}$) and log-volume ($H_{v,t}^{(i)}$) in aligned windows, and introduce the Coupling Intensity Index (CII), a rolling Pearson correlation that measures their temporal co-movement. Within-firm temporal coupling is strong and positive: mean $r = 0.531$ across the 488-firm panel, with 92.7% of firms exhibiting positive coupling and 80.4% above $r = 0.3$. The pattern holds across all eleven GICS sectors. The cross-sectional analogue at $G = 488$ shows a small but statistically detectable positive correlation between full-sample $H_{|r|}$ and H_v ($r = 0.18$, $p < 0.001$); within-firm temporal coupling is roughly threefold stronger and structurally different. Coupling heterogeneity rises sharply in high-VIX regimes (Mann-Whitney high vs. low $p = 0.0019$, with the upper-quartile distribution exhibiting both higher medians and substantially greater dispersion), but mean coupling does not uniformly intensify with stress. The predictive question has a more layered answer. Under inference robust to two-way clustering by firm and time (Cameron et al., 2011), Bell–McCaffrey CR2 with Satterthwaite degrees of

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freedom (Imbens and Kolesár, 2016), and the wild cluster restricted bootstrap (MacKinnon and Webb, 2017), CII has no firm-specific incremental predictive content for any of the ten forward outcomes tested in this paper, including standard dollar-volume Amihud (Amihud, 2002) illiquidity, the Corwin–Schultz spread (Corwin and Schultz, 2012), the Roll spread (Roll, 1984), realised volatility, max drawdown, abnormal turnover, vol-of-vol, log Amihud, Δ Amihud, and forward Corwin–Schultz / Roll spreads. Inference methods that condition on the time dimension instead (time-clustered standard errors, Driscoll–Kraay HAC) detect a significant cross-firm time-period co-movement that the firm-conditional methods do not. The natural reading is that CII inherits a market-wide stress signal that moves with forward liquidity proxies in aggregate but adds no firm-specific predictive content beyond what the standard liquidity-and-volatility toolkit already captures; every focal-only signal in the battery is fully absorbed by nine standard benchmarks in the combined horse-race specification. An apparently strong predictive result against a non-standard share-volume illiquidity proxy dissipates once both proper Amihud convention and dependence-robust inference are imposed; I document and dissect this as a price-level confound. The contribution is therefore a strong and replicable within-firm fractal coupling phenomenon, a sharp absorption pattern that pinpoints the dimension of dependence carrying CII’s apparent signal, and three methodological cautions for the wider econophysics-meets-finance literature: distinguish share-volume from dollar-volume liquidity proxies, report inference at the dimension that matches the unit of analysis claimed, and treat the Bell–McCaffrey Satterthwaite df as a routine diagnostic rather than assuming it scales with cluster count.

Keywords: Hurst exponent, Detrended fluctuation analysis, Price-volume relationship, Long memory, Coupling Intensity Index, Amihud illiquidity, Cluster-robust inference, Wild cluster bootstrap, Methodological caveat

1. Introduction

Financial price volatility and trading volume each exhibit long-range dependence.² Absolute returns display Hurst exponents of approximately 0.7–

²This paper builds on an earlier MSc project (Malempati Hari, 2013) in which I examined whether Hurst-based fractal characteristics of stock-price series were associated with

0.8 (Ding et al., 1993; Bollerslev and Jubinski, 1999), while volume persists even more strongly at $H \approx 0.7\text{--}0.9$ (Lobato and Velasco, 2000). Whether these two fractal regimes are *coupled* (whether they co-evolve over time, and whether that co-evolution carries information a financial economist would care about) has not been systematically investigated.

This paper addresses both questions. I distinguish between *static* coupling (the cross-sectional correlation of Hurst exponents across firms) and *temporal* coupling (the within-firm co-movement of rolling Hurst exponents over time), and I construct a Coupling Intensity Index (CII) that measures the strength of temporal coupling as a time-varying quantity. I then test four hypotheses:

- H1** *Static coupling:* Firms with more persistent volume also have more persistent volatility in the cross-section.
- H2** *Temporal coupling:* Rolling Hurst exponents for $|r_t|$ and $\log V_t$ co-move positively within firms over time.
- H3** *Regime dependence:* Temporal coupling intensifies during high-uncertainty regimes (elevated VIX).
- H4** *Predictive content:* CII has incremental predictive content for standard forward liquidity and volatility measures.

The results are as follows. H1 is reframed rather than rejected: the cross-sectional Pearson correlation between full-sample $H_{|r|}$ and H_v at $G = 488$ is small but statistically detectable ($r = 0.18$, $p < 0.001$). The original v1.0 50-firm sample returned a null on the same construction; the broader panel surfaces a modest cross-sectional dependence that the smaller sample lacked the power to detect. H2 is strongly supported at the full S&P 500 scale and is roughly threefold stronger than the cross-sectional analogue: mean within-firm temporal coupling is $r = 0.531$, with 92.7% of the 488 panel firms exhibiting positive coupling and 80.4% above $r = 0.3$; the pattern is

average trading volume in the cross section. That earlier work found a weak but statistically significant static relationship. The present study extends that line of inquiry by shifting from static association to temporal co-evolution, asking whether the persistence structures of volatility and volume move together through time and whether that coupling predicts future market illiquidity.

universal across all eleven GICS sectors. The interesting empirical object remains the within-firm temporal coupling, not the modest cross-sectional level dependence. H3 is confirmed in a sharper form than its original 50-firm version: a Mann-Whitney test for the high-VIX vs. low-VIX coupling distribution rejects the null at $p = 0.0019$, but the rejection is driven by distributional shape rather than mean shift. The high-VIX regime exhibits a higher median of within-firm coupling and substantially greater cross-firm dispersion; mean coupling does not uniformly intensify with stress.

H4, by contrast, is rejected for firm-specific predictive content but exhibits a structural inference disagreement that is itself informative. Under inference methods that condition on the firm dimension (firm-clustered, two-way clustered with the Cameron–Gelbach–Miller eigenvalue PSD adjustment, Bell–McCaffrey CR2, and the wild cluster restricted bootstrap), CII has no incremental predictive content for forward Amihud illiquidity computed in the standard dollar-volume convention of Amihud (2002), nor for realised volatility, the Corwin–Schultz spread, the Roll spread, the volatility of realised volatility, the maximum drawdown, abnormal turnover, log Amihud, or Δ Amihud. Inference methods that condition on the time dimension instead (time-clustered standard errors, Driscoll–Kraay HAC) detect a statistically significant cross-firm time-period co-movement between CII and several forward liquidity proxies. The horse race against nine standard liquidity benchmarks is decisive: every focal-only signal in the battery is fully absorbed in the combined specification (the strongest case is the forward Corwin–Schultz spread, where focal two-way $t = +3.64$ collapses to combined $t = +0.07$). The reading is that CII inherits a market-wide stress co-movement that moves with several forward liquidity proxies in aggregate but adds no firm-specific predictive content beyond what the standard toolkit already captures.

A specification that uses share volume rather than dollar volume in the Amihud denominator does produce a statistically significant CII coefficient at $G = 50$ and survives the wild cluster restricted bootstrap. That signal disappears once dollar volume replaces share volume in the denominator, as standard practice requires. Section 5.5.5 dissects the divergence as a price-level confound: the share-volume Amihud co-moves with within-firm price-level drift, and CII correlates with that drift component rather than with price impact per dollar traded. The fix is in the dependent-variable definition rather than in the standard errors, and modern cluster-robust inference (including the wild cluster bootstrap) does not detect it.

This combination of results is sharper than the broad predictive thesis the paper originally set out to test. Prior work on price-volume fractal relationships (Podobnik et al., 2009) and common long-memory components (Bollerslev and Jubinski, 1999) has established that volatility and volume share structural features. The present paper shows that the within-firm coupling of those structural features (a) exists in a strong and quantitatively measurable form across the full S&P 500, (b) is regime-dependent in a heterogeneity-amplifying rather than mean-amplifying sense, (c) carries no firm-specific predictive content for standard market-quality outcomes once the firm dimension of dependence is properly modelled, and (d) co-moves with cross-firm time-period stress dynamics in ways that several inference methods detect but that horse-race conditioning fully absorbs. The empirical claim is narrower than the original H4 but is structurally informative: the inference-method disagreement pinpoints exactly which dimension of dependence carries the apparent signal and which does not. The paper also delivers two transferable cautionary lessons: distinguish share-volume from dollar-volume illiquidity proxies, and report inference at the dimension that matches the unit of analysis the paper actually claims, since the choice can determine whether one finds a signal or a null at the same data.

The conceptual distinction that organises the paper is between first-order and second-order fractal properties. Detrended cross-correlation analysis (DCCA; Podobnik and Stanley, 2008) measures whether price and volume fluctuations are correlated at different scales, a first-order property. The rolling Hurst approach measures whether the *scaling exponents themselves* (which summarise how memory decays) evolve in tandem. This is a second-order property: a statement about the dynamics of complexity rather than the dynamics of levels.

2. Literature

2.1. Long Memory in Returns and Volatility

Mandelbrot (1963) first demonstrated that financial prices exhibit non-Gaussian, scale-invariant properties. The Hurst exponent H , originally developed for hydrological applications (Hurst, 1951) and formalised through fractional Brownian motion (Mandelbrot and Van Ness, 1968), provides the standard measure of long-range dependence. Cont (2001) identified the slow decay of autocorrelation in absolute returns as among the most robust stylised facts in finance. Ding et al. (1993) showed that this long memory is strongest

for $|r_t|^d$ at $d \approx 1$, persisting over hundreds of lags. Lo (1991) cautioned that naive R/S analysis can overstate long memory when short-range dependence is present, motivating the use of detrended fluctuation analysis (Peng et al., 1994). Finite-sample properties and heavy-tail biases have been characterised by Weron (2002) and Barunik and Kristoufek (2010). The HAR-RV model of Corsi (2009) provides an influential parsimonious framework for realised volatility forecasting, exploiting the heterogeneous autoregressive structure of realised measures.

2.2. Long Memory in Trading Volume

Lobato and Velasco (2000) provided rigorous evidence for fractional integration in volume, estimating $H \approx 0.7$ – 0.9 across NYSE equities. This finding is even more robust than the volatility long-memory result and has been replicated across markets and frequencies. Plerou et al. (2003) documented power-law correlations between price changes and volume, while Ray and Tsay (2000) confirmed long-range dependence in daily stock volatilities. Chordia et al. (2001) showed that trading activity predicts expected stock returns through its relationship to liquidity provision.

2.3. Liquidity, Volume, and Market Microstructure

The Amihud illiquidity ratio (Amihud, 2002) has become a standard measure of price impact per unit of trading volume, capturing the adverse-selection component of illiquidity. Kyle (1985) provided the theoretical foundation for relating information asymmetry to market depth. Brunnermeier and Pedersen (2009) formalised the feedback loop between volatility and liquidity: when volatility rises, margins tighten, forcing deleveraging that further reduces liquidity. This volatility-liquidity spiral provides a theoretical channel through which fractal coupling between volatility persistence and volume persistence could predict liquidity deterioration.

2.4. Joint Dynamics: The MDH and Beyond

The Mixture of Distributions Hypothesis (Clark, 1973; Tauchen and Pitts, 1983; Epps and Epps, 1976) posits a latent information arrival process that simultaneously governs volatility and volume. Andersen (1996) recast the MDH within a stochastic volatility framework, showing time-varying information flow generates the observed co-clustering. Bollerslev and Jubinski (1999) identified a common long-memory component shared by volatility and volume using ARFIMA methods. Podobnik et al. (2009) applied

DCCA to measure scale-dependent cross-correlations between price changes and volume changes. Karpoff (1987) and Gallant et al. (1992) established the broader empirical price-volume relationship. Kristoufek (2015) reviewed the methodological challenges and open problems in measuring power-law cross-correlations, including bias and finite-sample issues that motivate the rolling-window approach adopted here.

Several studies have computed time-varying Hurst exponents for individual series. Cajueiro and Tabak (2004) examined emerging-market efficiency. Alvarez-Ramirez et al. (2008) characterised U.S. stock markets through erratic dynamics of the rolling Hurst exponent estimated by DFA. Grech and Mazur (2004) and Grech and Pamuła (2008) used the local Hurst exponent to study crash dynamics on the Polish stock exchange. Garcin (2017) developed a variational-smoothing estimator for time-dependent Hurst exponents and applied it to foreign-exchange rate forecasting, providing the closest existing methodological cousin to the rolling-window approach adopted here. Antoniadis et al. (2021) introduced a visual warning tool based on time-dependent generalised Hurst exponents H_q computed in overlapping windows, which is the broader framework into which the rolling DFA construction used in this paper falls as a special case at $q = 2$. Vogl (2023) applied wavelet-denoised rolling Hurst exponents to S&P 500 returns in a nonlinear-dynamics framework. Brouty and Garcin (2024) linked entropy-based market information to the Hurst exponent under fractional Brownian motion and its Lamperti-stationary extension, providing a complementary information-theoretic framing to the present coupling-based approach. Garcin and Grasselli (2022) studied the rough-volatility dilemma across long and short time scales using a large dataset of stock indices and FX rates, documenting finite-sample bias issues in Hurst estimation that motivate the multi-estimator robustness check we report in Section 6. None of these studies, however, examines whether Hurst exponents computed from *two distinct series* co-move through time within a firm, nor whether such co-movement carries predictive content for standard market-quality outcomes. This is the gap the present study addresses.

3. Methodology

3.1. Detrended Fluctuation Analysis

For a time series $\{x_t\}_{t=1}^N$, the cumulative deviation profile is $Y(k) = \sum_{t=1}^k (x_t - \bar{x})$. The profile is divided into non-overlapping windows of length

s , each detrended by a fitted polynomial of order m (here $m = 1$, i.e., DFA-1). The fluctuation function is:

$$F(s) = \left[\frac{1}{N_s} \sum_{\nu=1}^{N_s} \frac{1}{s} \sum_{k=1}^s \left(Y((\nu-1)s+k) - \tilde{Y}_\nu(k) \right)^2 \right]^{1/2}. \quad (1)$$

For a long-range correlated process, $F(s) \sim s^H$. The Hurst exponent H is estimated as the ordinary least squares slope of $\log F(s)$ versus $\log s$. The scale range is $s_{\min} = 10$ to $s_{\max} = \lfloor N/4 \rfloor$, with logarithmic spacing using a multiplicative factor of 1.2 (i.e., scales $s_k = \lfloor s_{\min} \times 1.2^k \rfloor$ for $k = 0, 1, 2, \dots$ up to $s_{\max}$). Fit quality is assessed via R^2 of the log-log regression.

3.2. Series Construction

For each equity i , three series are constructed from daily close prices $P_t^{(i)}$ and volumes $V_t^{(i)}$:

$$r_t^{(i)} = \log P_t^{(i)} - \log P_{t-1}^{(i)}, \quad (2)$$

$$|r_t^{(i)}| = |\log P_t^{(i)} - \log P_{t-1}^{(i)}|, \quad (3)$$

$$v_t^{(i)} = \log(V_t^{(i)} + 1). \quad (4)$$

Stationarity is verified via the ADF and KPSS tests. All log-return series are confirmed stationary.

3.3. Rolling Hurst Exponents

The rolling Hurst exponent at time t for window length W and step size Δ is defined as:

$$H_{|r|,t}^{(i)} = \text{DFA}(\{|r_\tau^{(i)}|\}_{\tau=t-W+1}^t), \quad H_{v,t}^{(i)} = \text{DFA}(\{v_\tau^{(i)}\}_{\tau=t-W+1}^t), \quad (5)$$

where $H_{|r|,t}^{(i)}$ denotes the Hurst exponent of absolute returns (the volatility proxy) and $H_{v,t}^{(i)}$ denotes the Hurst exponent of log-volume for firm i at rolling index t . The primary specification uses $W = 500$ trading days and $\Delta = 20$ days, with sensitivity checks at $W \in \{252, 500, 756\}$.

3.4. Coupling Metrics

Static coupling is the cross-sectional Pearson correlation of full-sample $H_{|r|}^{(i)}$ and $H_v^{(i)}$ computed across all $i = 1, \dots, 488$ firms in the Phase 2 panel.

Temporal coupling for firm i is the Pearson correlation computed over the full set of rolling indices:

$$\rho^{(i)} = \text{Corr}\left(H_{|r|,t}^{(i)}, H_{v,t}^{(i)}\right), \quad t \in \mathcal{T}^{(i)}, \quad (6)$$

where $\mathcal{T}^{(i)}$ is the discrete set of rolling-window evaluation dates for firm i .

Coupling Intensity Index (CII). The CII is a time-varying measure defined as the sample Pearson correlation computed over a trailing window of L rolling-Hurst observations. Formally, let $\mathcal{T}_{t,L} = \{t_j \in \mathcal{T}^{(i)} : t_j \leq t, |\{t_k \in \mathcal{T}^{(i)} : t_k \leq t\}| - j < L\}$ denote the L most recent rolling-Hurst evaluation dates up to and including calendar day t . Then:

$$\text{CII}_t^{(i)} = \frac{\sum_{j \in \mathcal{T}_{t,L}} (H_{|r|,j}^{(i)} - \bar{H}_{|r|}^{(i)}) (H_{v,j}^{(i)} - \bar{H}_v^{(i)})}{\sqrt{\sum_{j \in \mathcal{T}_{t,L}} (H_{|r|,j}^{(i)} - \bar{H}_{|r|}^{(i)})^2 \sum_{j \in \mathcal{T}_{t,L}} (H_{v,j}^{(i)} - \bar{H}_v^{(i)})^2}}, \quad (7)$$

where $\bar{H}_{|r|}^{(i)}$ and $\bar{H}_v^{(i)}$ are the sample means over $\mathcal{T}_{t,L}$. The primary specification uses $L = 30$ rolling-Hurst observations (approximately 600 trailing trading days given $\Delta = 20$). Sensitivity to $L \in \{20, 30, 40\}$ is reported below.

Timing convention. $\text{CII}_t^{(i)}$ is computable at the close of trading day t and uses only data through day t . All predictive outcome variables are measured over the strictly future window $[t + 1, t + h]$. No observations overlap between the CII estimation window and the outcome measurement window.

3.5. Predictive Regression

To test H4, I estimate panel regressions with firm fixed effects:

$$Y_{t+h}^{(i)} = \alpha_i + \beta_1 \text{CII}_t^{(i)} + \beta_2 H_{|r|,t}^{(i)} + \beta_3 H_{v,t}^{(i)} + \varepsilon_t^{(i)}, \quad (8)$$

where $Y_{t+h}^{(i)}$ is a forward-looking outcome measured over $[t + 1, t + h]$:

- *Amihud illiquidity* (Amihud, 2002), dollar-volume convention scaled by 10^6 : $\text{ILLIQ}_{t+h}^{(i)} = \frac{10^6}{h} \sum_{\tau=t+1}^{t+h} \frac{|r_\tau^{(i)}|}{V_\tau^{(i)} \cdot P_\tau^{(i)}}$

- *Realised volatility*: $RV_{t+h}^{(i)} = \sqrt{\sum_{\tau=t+1}^{t+h} (r_{\tau}^{(i)})^2}$
- *Abnormal turnover*: forward share volume normalised by trailing 60-day mean
- *Maximum drawdown*: largest peak-to-trough decline in $[t+1, t+h]$

The dollar-volume denominator $V_{\tau}^{(i)} \cdot P_{\tau}^{(i)}$ is the standard Amihud convention. An earlier version of this study used share volume in the denominator and produced a statistically significant CII coefficient that does not survive the standard convention; Section 5.5.5 treats this as a substantive finding in its own right.

Amihud illiquidity is the *primary* H4 outcome of the paper. The remaining three (realised volatility, abnormal turnover, maximum drawdown) and the further proxies introduced in Section 5.5 (Corwin–Schultz spread, Roll spread, vol-of-vol, log Amihud, and forward Δ Amihud) are reported as *exploratory secondary outcomes*. Because all ten outcomes return statistically null estimates for the focal β_1 coefficient, family-wise multiple-testing corrections would only widen the reported p -values further from significance and are omitted as redundant. Horizons are $h \in \{5, 21\}$ trading days (one week, one month).

Because panel observations are likely correlated both within firms (persistence) and within time periods (common shocks), I report five baseline standard-error specifications: heteroskedasticity-consistent (HC1), clustered by firm, clustered by time, two-way clustered by firm and time following Cameron et al. (2011), and Newey–West with automatic bandwidth selection (Newey and West, 1987). The two-way clustered specification accounts for both within-firm persistence and cross-sectional correlation from common shocks. For the combined horse-race specification (Section 5.5.3) I additionally report two small-sample corrections recommended in modern cluster-robust practice: the Bell–McCaffrey CR2 standard error with Satterthwaite effective degrees of freedom (Imbens and Kolesár, 2016), and the wild cluster restricted bootstrap with Rademacher weights following Cameron et al. (2008) and MacKinnon and Webb (2017). These provide stricter inference than the asymptotic CRVE at $G = 50$ firms; details are in Section 5.5.3 and Table 5. The coefficient of interest is β_1 : does CII predict future outcomes *beyond* what the individual Hurst levels already capture?

3.6. Regime Conditioning

To test H3, the CBOE Volatility Index (VIX) is classified into quartile-based categories: low (< 25 th percentile), medium (25th–75th), and high (> 75 th percentile). Temporal coupling $\rho^{(i)}$ is computed separately within each VIX category, and differences are tested via the Mann-Whitney U statistic.

3.7. Robustness

Eleven checks are conducted: window sensitivity ($W \in \{252, 500, 756\}$), non-overlapping windows, first-differenced Hurst series, shuffled surrogates, squared returns as alternative volatility proxy, subperiod analysis, sector decomposition, market factor control (partial correlation after removing SPY Hurst dynamics), DFA polynomial order (linear vs. quadratic), alternative Hurst estimators (R/S, MFDFA at $q = 2$), and Granger causality between the two Hurst series. The Granger causality tests are exploratory: given the irregular spacing and overlapping-window construction, they should be interpreted as directional descriptives rather than formal causal evidence.

3.8. Pre-Registration of the Phase 2 Universe Expansion

The empirical results in this paper come in two layers. The first layer is the v1.0–v1.2 50-firm large-cap analysis that introduced the Coupling Intensity Index, documented the static-temporal coupling distinction, identified the share-volume vs. dollar-volume Amihud confound, and reported the predictive null for CII against the standard dollar-volume Amihud convention. The second layer is the Phase 2 universe expansion to $G = 488$ S&P 500 firms. The Phase 2 expansion was executed under a pre-registration protocol committed to the public repository before any $G > 50$ data fetch.

The protocol document, archived as `research/horse_race/PHASE2_PROTOCOL.md` at master HEAD `f8aa564`, specified in advance: the universe definition (current April 2026 S&P 500 constituency), the time window (2015-01-01 to 2026-04-01), the inclusion criteria (≥ 600 observations, ADF and KPSS stationarity, $\leq 5\%$ missing volume), the pipeline freeze (no code change between protocol freeze and Phase 2 run that affects estimated coefficients or inference), the ten target specifications (primary forward dollar-volume Amihud plus nine secondary outcomes), and a per-target decision rule that committed in advance to three outcome categories: (A) predictive null confirmed (twoway-cluster p AND Driscoll–Kraay p AND wild-cluster-bootstrap p all > 0.10); (B) combined-spec robust signal (twoway-cluster p AND DK p AND WCR p all < 0.05 with CR2 $p < 0.10$ in the combined-with-9-benchmarks

specification, plus same-sign and magnitude $\geq 25\%$ of the focal-only estimate); and (C) mixed inference (anything between A and B). The protocol explicitly disallowed adding new targets, changing window parameters, swapping inference methods, or reframing a Category C outcome as Category B after the run.

This pre-registration discipline matters for the inference disagreement reported in Section 5.5. The structural firm-conditional vs. time-conditional pattern was not selected ex post from a larger menu of possible inference choices. The six baseline standard-error specifications and the two small-sample corrections were committed in advance, run on $G = 488$ in a single execution, and reported in their entirety. The same applies to the per-target classification (Section 5 reports the protocol’s mechanical outcomes). Replication runs of `run_phase2.py` from the same commit produce identical numbers up to the WCR bootstrap seed, which is fixed in the repository.

4. Data

The sample comprises the 488 S&P 500 constituents (out of 503 attempted) for which Yahoo Finance returned at least 600 daily observations and the log-return series passed both ADF ($p < 0.01$) and KPSS ($p > 0.10$) stationarity tests over January 2, 2015 through April 1, 2026 ($\approx 2,800$ observations per equity). The universe spans all eleven GICS sectors. Constituency is current as of April 2026; the resulting sample inherits a survivorship-bias profile typical of contemporary S&P 500 universes and shared with most prior work using the same construction. VIX data are obtained from the CBOE via Yahoo Finance.

An earlier version of this study used a smaller 50-firm sample of large-cap S&P 500 names (v1.0–v1.2). The 488-firm panel reported here was added to test whether the small-sample inference observed at $G = 50$ was a power constraint or a robust feature of the data; that question is addressed in Section 5.5.3.

5. Results

5.1. Baseline Hurst Estimates

Across the 488-firm Phase 2 panel, $H_{|r|}^{(i)}$ averages 0.796 ($\sigma = 0.066$) and $H_v^{(i)}$ averages 0.879 ($\sigma = 0.073$), replicating the canonical long-memory findings on absolute returns and log-volume. The corresponding numbers in the

v1.0–v1.2 50-firm sample (Figure 1) are 0.795 and 0.881; the cross-firm mean is essentially unchanged when the universe expands tenfold. $H(\text{returns})$ in the v1.0 sample averages 0.461, confirming approximate market efficiency, and the 488-firm panel reproduces this near-0.5 level. Log-log fit quality of the DFA regressions is excellent (mean $R^2 = 0.994$; Figure 2). Bootstrap confidence intervals at the v1.0 sample confirmed $H \neq 0.5$ at $p < 0.001$ on both volatility and volume; the 488-firm cross-section produces tighter intervals at the same level of significance.

5.2. *H1: Static Coupling (Weak but Detectable at $G = 488$)*

The cross-sectional Pearson correlation between full-sample $H_{|r|}^{(i)}$ and $H_v^{(i)}$ across the 488-firm panel is $r = 0.183$ ($t = 4.11$, $p < 0.001$, $n = 488$). Spearman $\rho = 0.163$ ($p = 0.0003$). The original v1.0 50-firm sample reported a null at $r = -0.02$ ($p = 0.88$; Figure 5); the 488-firm cross-section is large enough to detect a small but statistically significant positive cross-sectional dependence that the smaller large-cap sample could not. **The H1 narrative therefore needs reframing rather than rejection.** The cross-sectional relationship is real but small: knowing a firm’s $H_{|r|}^{(i)}$ explains roughly $r^2 \approx 3.4\%$ of the cross-sectional variation in its $H_v^{(i)}$. The temporal coupling reported below (mean within-firm $r = 0.531$) is roughly threefold stronger and structurally different. The interesting empirical object remains within-firm co-evolution of memory structure, not the modest cross-sectional level dependence; the latter is consistent with shared sectoral or market-cap drivers of full-sample persistence levels and does not require the dynamic information-flow story that motivates CII.

5.3. *H2: Temporal Coupling (Confirmed)*

Across the 488-firm S&P 500 panel, the mean within-firm temporal Pearson correlation is $\bar{\rho} = 0.531$, with median 0.615, 92.7% of firms exhibiting positive coupling, 80.4% above $r = 0.3$, and 64.5% above $r = 0.5$. The minimum is $r = -0.570$ and the maximum is $r = 0.974$. **H2 is strongly confirmed.**

Sector decomposition shows the phenomenon is universal across the eleven GICS sectors: Energy leads at mean $r = 0.660$ (95.5% positive), followed by Consumer Discretionary (0.626, 95.8%), Financials (0.610, 94.7%), Health Care (0.543, 94.6%), Consumer Staples (0.541, 97.1%), Industrials (0.530, 92.2%), Communication Services (0.476, 87.0%), Information Technology

(0.471, 91.5%), Real Estate (0.461, 87.1%), Materials (0.425, 92.3%), and Utilities (0.412, 87.1%). Every sector has at least 87% of firms with positive coupling.

The original 50-firm sample (large-cap focus, v1.0–v1.2) reported a slightly stronger mean ($\bar{\rho} = 0.665$, 49 of 50 firms positive). The $G = 488$ mean is closer to the lower end of the cross-sector distribution because the broader universe includes mid-cap and recently-added firms with lower coupling magnitudes. The phenomenon is strongest in large-cap names but holds across the full S&P 500 constituency.

Figure 3 illustrates the co-movement for AAPL ($r = 0.759$), JPM ($r = 0.901$), and META ($r = -0.421$, the most extreme negative case in the original 50-firm sample, coinciding with the firm’s 2022–23 corporate restructuring). The lead-lag profile (Figure 6) peaks at lag zero and decays symmetrically, consistent with a shared contemporaneous driver.

Exploratory Granger causality tests on the 50-firm subsample revealed a directional asymmetry: volume persistence Granger-causes volatility persistence in 26% of tickers versus 8% in the reverse direction. While this three-fold asymmetry is suggestive of information manifesting first in trading activity (Gallant et al., 1992; Kyle, 1985), the overlapping-window construction limits the strength of causal claims from these tests; the result is reported as exploratory and was not re-run on the full 488-firm panel.

5.4. *H3: Regime Dependence (Confirmed in Distribution Shape)*

Temporal coupling conditioned on VIX quartile-based categories at $G = 488$:

VIX Category	Mean ρ	Median ρ	Std	n firms
Low (< 25th pctl)	0.567	0.666	0.344	495
Medium (25th–75th)	0.497	0.566	0.299	495
High (> 75th pctl)	0.563	0.738	0.450	495

Table 1: Temporal coupling by VIX regime, $G = 488$. Mann-Whitney test (high > low) on the cross-firm distributions: $U = 135,542$, $p = 0.0019$.

H3 is confirmed. Coupling is significantly higher during high-VIX periods. The subperiod analysis reinforces this: coupling nearly doubles from the pre-COVID baseline ($r = 0.41$) to the crisis period ($r = 0.77$; Figure 7), then attenuates post-crisis ($r = 0.30$).

5.5. *H4: Predictive Content (Firm-Specific Null, Time-Period Co-Movement)*

H4 asks whether CII has incremental predictive content for standard forward liquidity and volatility measures. The answer, after running ten target specifications across six baseline standard-error methods plus two small-sample corrections at $G = 488$, is structured: under inference methods that condition on the firm dimension (firm-clustered, two-way clustered, Bell–McCaffrey CR2 with Satterthwaite df, the wild cluster restricted bootstrap), the CII coefficient is statistically null for every target in every specification. Under inference methods that condition on the time dimension instead (time-clustered standard errors, Driscoll–Kraay HAC), several focal-only signals reach the 5% threshold for the primary outcome and for five of the eight secondary outcomes. The horse race against nine standard liquidity benchmarks is decisive: every focal-only signal in the battery is fully absorbed in the combined specification, regardless of which SE method is used.

The reading these results support is that CII inherits a market-wide stress co-movement that affects multiple forward liquidity proxies in aggregate, but adds no firm-specific predictive content beyond what the standard liquidity-and-volatility toolkit already captures. The inference disagreement pinpoints exactly which dimension of dependence carries the apparent signal: the cross-firm time-period dimension, not the firm-specific dimension. Section 5.5.5 below additionally explains why an earlier version of this study using share-volume Amihud reported a robust positive result: it was driven by a price-level component of the non-standard denominator, not by predictive content for price impact per dollar traded. Section 5.5.4 addresses the third methodological observation of this paper: the Bell–McCaffrey CR2 Satterthwaite effective degrees of freedom collapsed to 1.0 at $G = 488$, which carries practical implications for cluster-robust inference in panels with leverage-heterogeneous firms.

5.5.1. *Primary Specification: Forward Amihud Illiquidity*

Table 2 reports panel regression results for Equation (8) at the one-month horizon ($h = 21$) on the 488-firm panel. The dependent variable is the standard Amihud illiquidity defined in Section 3.5 with dollar volume in the denominator.

Table 2: Inference for $\text{CII} \rightarrow \text{standard (dollar-volume) Amihud illiquidity}$, $G = 488$, focal specification with Hurst controls ($h = 21$ trading days).

SE Method	t -statistic	p -value	df
HC1 (White)	-2.56	0.011	41,569
Firm-clustered	-0.92	0.357	487
Time-clustered	-2.61	0.011	83
Two-way clustered	-0.93	0.356	83
Newey–West	-1.48	0.138	41,569
Driscoll–Kraay	-2.25	0.025	872
Bell–McCaffrey CR2	-0.91	0.527	$\text{df}_{\text{satt}} = 1.02$
WCR bootstrap	—	0.720	$n_{\text{boot}} = 999$

Panel regressions with firm fixed effects, $N = 41,572$ observations across 488 equities and 875 calendar months. Regressors: $\text{CII}_t^{(i)}$, $H_{|r|,t}^{(i)}$, $H_{v,t}^{(i)}$. Two-way clustering follows Cameron et al. (2011) with the eigenvalue PSD adjustment of CGM (2011) §2.3 applied when needed. Driscoll–Kraay uses Bartlett bandwidth ≥ 21 (Driscoll and Kraay, 1998). CR2 follows Imbens and Kolesár (2016) at the firm dimension; WCR follows Cameron et al. (2008) and MacKinnon and Webb (2017) with Rademacher weights and the restricted null. Bold rows mark methods that detect a significant CII coefficient at the 5% level.

The CII coefficient is $\hat{\beta}_{\text{CII}} = -2.6 \times 10^{-3}$. The disagreement between firm-conditional methods (firm-clustered, two-way, CR2, WCR) and time-conditional methods (time-clustered, Driscoll–Kraay) is structural rather than ad hoc. Methods that allow within-firm correlation but not within-time correlation see a null because the apparent signal lives in the cross-firm time-period dimension. Methods that allow within-time correlation see a t near -2.5 because that cross-firm dimension is real. The two-way clustered specification, which allows both, attributes the shared variance to the larger of the two cluster sets and ends up null. The natural interpretation: the CII coefficient on forward Amihud reflects co-movement of CII and Amihud at the cross-firm time-period level (a market-wide stress dynamic), not within-firm one-month-ahead forecasting power. Whether one calls this “predictive content” is a matter of definition; under the standard asset-pricing definition,

where the unit of forecast is firm-month, the result is null.

5.5.2. Battery of Standard Liquidity and Volatility Outcomes

Table 3 extends the test to nine target variables (the primary plus eight secondary outcomes), each in both focal and combined specifications.

Table 3: Predictive battery, $G = 488$: CII coefficient and inference for nine target specifications, focal and combined-with-9-benchmarks regressions, $h = 21$.

Target	$\hat{\beta}_{\text{CII}}^{\text{focal}}$	$t_{2\text{way}}^{\text{focal}}$	$t_{\text{DK}}^{\text{focal}}$	$t_{2\text{way}}^{\text{combined}}$	$t_{\text{DK}}^{\text{combined}}$
<i>Standard Amihud and transformations</i>					
Amihud illiquidity (PRIMARY)	-2.6×10^{-3}	-0.93	-2.25*	-0.77	-1.75
log Amihud	+0.071	+2.19*	+1.92	+0.21	+0.21
Δ Amihud	-4.4×10^{-4}	-1.61	-0.56	-0.96	-0.97
<i>Spread-based liquidity proxies</i>					
Forward Corwin–Schultz spread	$+5.1 \times 10^{-4}$	+3.64**	+3.45**	+0.07	+0.10
Forward Roll spread	$+1.2 \times 10^{-3}$	+2.07*	+1.99*	+0.62	+0.76
<i>Other forward outcomes</i>					
Forward vol-of-vol	$+4.7 \times 10^{-3}$	+1.40	+1.13	-1.11	-1.87
Realised volatility	$+6.7 \times 10^{-3}$	+2.49*	+2.33*	+0.08	+0.10
Maximum drawdown	-6.3×10^{-3}	-1.74	-1.83	-0.81	-0.92
Abnormal turnover	+0.254	+1.03	+1.01	+1.08	+1.04

Panel regressions with firm fixed effects. Focal regressors: $\text{CII}_t^{(i)}$, $H_{|r|,t}^{(i)}$, $H_{v,t}^{(i)}$. Combined adds the nine standard benchmarks listed in Table 4. Two-way clustered SEs (Cameron–Gelbach–Miller 2011) and Driscoll–Kraay SEs (Bartlett bandwidth ≥ 21) are reported. Bold cells mark $p < 0.05$. * $p < 0.05$; ** $p < 0.01$. The pattern is uniform: focal-only coefficients that reach significance for the primary outcome and for five of eight secondaries are absorbed to null in the combined specification across both SE methods. The forward Corwin–Schultz spread illustrates the absorption most cleanly ($t = +3.64 \rightarrow +0.07$, four-orders-of-magnitude drop in the coefficient).

Reading the table: at $G = 488$, several focal-only specifications cross the 5% threshold under two-way clustering or Driscoll–Kraay HAC. Every single focal-significant coefficient becomes statistically null in the combined specification. The implication is identical for all nine targets: CII has measurable contemporaneous co-movement with several forward liquidity and volatility proxies, but no incremental predictive content beyond what the

standard nine-benchmark toolkit already captures. The forward Corwin–Schultz spread is the most striking illustration: the coefficient drops by four orders of magnitude (from 5.1×10^{-4} to 1.1×10^{-5}) and the t -statistic falls from +3.64 to +0.07 once the standard benchmarks are added.

Multiple-testing handling. The nine-target battery scans nine outcomes; without correction, the family-wise probability of finding at least one significant focal-only t -statistic at the 5% level under the global null is roughly $1 - 0.95^9 \approx 0.37$. The horse-race absorption pattern is the binding multiple-testing handling for this family: every focal-only signal disappears in the combined-with-9-benchmarks specification, regardless of which target is tested. The combined specification is the relevant inference for the question “does CII add incremental predictive content beyond the standard liquidity-and-volatility toolkit,” and the combined specification is null on every target. As a redundant cross-check, applying the Bonferroni (1936) correction to the four primary outcomes from the 50-firm broader-outcome scan (Amihud illiquidity, realised volatility, max drawdown, abnormal turnover) at the focal-only level requires $p < 0.05/4 = 0.0125$ for family-wise significance; under two-way clustering at $G = 488$ no target meets this corrected threshold. The Romano–Wolf step-down procedure of Romano and Wolf (2005) would yield similar conclusions and is left for an appendix in a future revision.

5.5.3. Horse Race Against Standard Liquidity Predictors

A horse race against nine standard liquidity benchmarks (Roll spread, Corwin–Schultz MSPREAD₀, dollar-volume rolling Amihud, lagged realised volatility, lagged Amihud, vol-of-vol, normalised turnover, abnormal-turnover spike, and Cboe VIX), following the protocol of Goyenko et al. (2009) §5.2, confirms the same null. Table 4 reports CII alongside each benchmark in a per-benchmark regression and in a single combined specification.

Table 4: Horse race: CII against nine standard liquidity predictors for forward dollar-volume Amihud, $G = 488$, h

Specification	$\hat{\beta}$	t (twoway)	p (twoway)	t (DK)	p (DK)
<i>CII alone with Hurst controls (focal):</i>					
CII coefficient	-2.6×10^{-3}	-0.93	0.356	-2.25*	0.025
<i>Per-benchmark coefficient on the named benchmark (CII + benchmark + Hurst controls):</i>					
Roll (1984) spread	+0.042	+0.85	0.400	+1.95	0.050
Corwin–Schultz (2012) MSPREAD ₀	-0.286	-1.06	0.293	-1.31	0.187
Amihud (2002) rolling, dollar-volume	+0.237	+10.11**	<0.001	+4.41**	<0.001
Normalised turnover	-1.5×10^{-3}	-6.08**	<0.001	-3.06**	<0.001
Volatility of volatility	$+7.5 \times 10^{-3}$	+0.93	0.356	+2.41*	0.016
Lagged realised volatility	+0.026	+1.07	0.288	+3.67**	<0.001
Lagged Amihud illiquidity	+0.130	+2.67**	0.009	+2.50*	0.012
Abnormal-turnover spike	-2.3×10^{-3}	-2.42*	0.018	-1.82	0.070
Cboe VIX	$+1.7 \times 10^{-4}$	+1.06	0.292	+4.38**	<0.001
<i>All nine benchmarks plus CII (combined):</i>					
CII coefficient	-1.4×10^{-3}	-0.77	0.445	-1.75	0.080

Panel regressions with firm fixed effects, $N = 41,572$ observations across 488 equities and 875 months. Two-way clustering follows Cameron et al. (2011) with the eigenvalue PSD adjustment of CGM (2011) §2.3 and h when needed. Driscoll–Kraay uses Bartlett bandwidth ≥ 21 (Driscoll and Kraay, 1998). * $p < 0.05$; ** $p < 0.01$. The CII coefficient inside each per-benchmark specification (not shown in this table) is statistically null under two-way clustering in every row, confirming that CII does not survive horse-race conditions. The firm dimension regardless of which benchmark is the competitor; under Driscoll–Kraay HAC, the CII coefficient is significant in seven of nine per-benchmark rows, mirroring the focal-only firm-vs-time inference pattern. The combined specification absorbs CII to null under firm-conditional inference and to marginally null under Driscoll–Kraay. The dollar-volume rolling Amihud benchmark is the dominant predictor of forward dollar-volume Amihud as expected for a persistent variable; lagged Amihud, normalised turnover, and abnormal-turnover spike also contribute under firm-conditional inference; Driscoll–Kraay further surfaces volatility-of-volatility, lagged realised volatility, and the VIX. Benchmark estimators cite Roll (1984), Corwin and Schultz (2006), Amihud (2002), with convention recommendations from Goyenko et al. (2009), Hasbrouck (2009), and Chordia and Ranaldo (2017).

For realised volatility the same horse race produces a similar absorption: focal $t_{2\text{way}} = +2.49$ ($p = 0.015$, significant) collapses to combined $t_{2\text{way}} = +0.08$ ($p = 0.94$, fully null). Table 5 confirms both target results under modern small-sample corrections.

Table 5: Small-sample-corrected inference for CII in the combined specification at $G = 488$, dollar-volume Amihud and realised volatility targets.

Target	$\hat{\beta}_{\text{CII}}$	CR2 t	CR2 p	Satterthwaite df	WCR p
Amihud illiquidity (dollar-volume)	-1.4×10^{-3}	-0.85	0.544	1.07	0.725
Realised volatility	$+2.7 \times 10^{-4}$	+0.41	0.700	1.20	0.730

CR2 is the Bell–McCaffrey cluster-robust variance with Satterthwaite effective degrees of freedom (Imbens and Kolesár, 2016), computed at the firm dimension. WCR is the wild cluster restricted bootstrap with 999 Rademacher resamples under the null (Cameron et al., 2008; MacKinnon and Webb, 2017). Both confirm the null established by the firm-conditional cluster-robust tests. The Satterthwaite effective degrees of freedom collapsed from approximately 6 at $G = 50$ to approximately 1 at $G = 488$, despite the $9.8\times$ increase in cluster count; this is addressed in Section 5.5.4.

5.5.4. Diagnostic: Cluster Heterogeneity and the Satterthwaite df Collapse at $G = 488$

A striking feature of the small-sample inference at $G = 488$ is that the Bell–McCaffrey CR2 Satterthwaite effective degrees of freedom collapsed to approximately 1.0, despite the cluster count rising from 50 to 488. Naive intuition says that the asymptotic CRVE reference distribution at $t(G - 1) = t(487)$ should have replaced the more conservative small-sample distributions; in practice the effective df at the firm dimension barely moved, and remained smaller than at $G = 50$ in some specifications. The mechanism is well understood in the cluster-robust literature (Imbens and Kolesár, 2016): the Satterthwaite df is governed by the second moments of the cluster-level score contributions, not by the cluster count. When a small number of high-leverage firms (typically illiquid small-cap or recent-IPO names with extreme Amihud values) carry most of the residual variance, the meat matrix is dominated by those firms and the effective df does not scale with G .

The practical implication for the present paper is bounded: the CR2 result at $G = 488$ is consistent with the WCR bootstrap, the firm-clustered CRVE, and the two-way clustered CRVE, all of which agree on the firm-specific null for the CII coefficient. The CR2 inference adds no information here that the bootstrap and the simpler cluster-robust methods do not already provide.

The implication for the wider literature is more useful. Researchers running cluster-robust inference at $G \in [100, 500]$ in panels of U.S. equities should not assume that increasing cluster count is sufficient to make the asymptotic CRVE reliable. The Satterthwaite check should be reported

routinely as a diagnostic, and a sharply low effective df should be read as a flag that residual heterogeneity dominates and the bootstrap-based or randomization-based tests are the binding inference. The 50-firm vs. 488-firm comparison in this paper is a clean instance of this point.

5.5.5. Diagnostic: Price-Level Confound in the Share-Volume Amihud

An earlier version of this study used a non-standard share-volume illiquidity proxy as the dependent variable, computed as $\widetilde{\text{ILLIQ}}_{t+h}^{(i)} = h^{-1} \sum_{\tau} |r_{\tau}^{(i)}| / V_{\tau}^{(i)}$, with share volume rather than dollar volume in the denominator. Under that specification, CII appeared to be a strong predictor of forward illiquidity. Table 6 compares the two specifications side by side under identical regressors.

Table 6: Side-by-side comparison: share-volume vs. dollar-volume Amihud as dependent variable. Same focal regression and Hurst controls in both columns; only the dependent variable differs.

SE method	Share-volume $ r /V$ t	p	Dollar-volume $ r /(V \cdot P)$ t	p
<i>Focal regression with Hurst controls, $h = 21$:</i>				
HC1 (White)	9.67	<0.001	1.16	0.246
Firm-clustered	3.50	0.001	0.41	0.687
Time-clustered	4.53	<0.001	0.50	0.621
Two-way clustered	2.90	0.004	0.33	0.745
Newey–West	5.41	<0.001	0.56	0.579
Driscoll–Kraay	—	—	0.21	0.833
<i>Combined horse-race spec (CII + 9 benchmarks + Hurst controls):</i>				
WCR bootstrap	—	0.018	—	0.517
CR2 + Satterthwaite	—	0.066	—	0.507

The share-volume column reproduces the result reported in an earlier version of this paper and in the v1.1.0 SSRN preprint. The dollar-volume column is the paper’s primary specification. Driscoll–Kraay was not computed for the share-volume case.

The arithmetic relationship between the two proxies is

$$\widetilde{\text{ILLIQ}}_{\tau}^{(i)} = \frac{|r_{\tau}^{(i)}|}{V_{\tau}^{(i)}} = \frac{|r_{\tau}^{(i)}|}{V_{\tau}^{(i)} \cdot P_{\tau}^{(i)}} \cdot P_{\tau}^{(i)} = \text{ILLIQ}_{\tau}^{(i)} \cdot P_{\tau}^{(i)}. \quad (9)$$

The share-volume proxy is the standard dollar-volume Amihud multiplied by the contemporaneous price level. If CII at time t correlates with future price-level dynamics within a firm, then the share-volume proxy inherits that correlation through the multiplicative $P_\tau^{(i)}$ factor, even when CII is uncorrelated with price impact per dollar traded.

The data support exactly this interpretation. CII does not predict standard dollar-volume Amihud (Tables 2, 3, 4). It also does not predict realised volatility, forward Corwin–Schultz spread, or any of the other dollar-scaled liquidity measures examined. But within-firm price-level drift over multi-month windows is sufficient for the $P_\tau^{(i)}$ multiplier to inflate apparent illiquidity in firms or periods where prices are falling, and for that to align with periods of high CII via the same channel of fractal coupling and stress amplification that H3 confirms. The result is a positive cross-section of $\widetilde{\text{ILLIQ}}$ on CII that vanishes the moment P is divided out of the denominator.

This finding has two implications. The first is parochial: the original H4 hypothesis in this study is rejected once the standard convention is applied. The second is more useful to the wider literature, where the choice of denominator in price-impact proxies is the subject of an active and unresolved debate.

Lou and Shu (2017) decompose the Amihud measure and argue that its return premium is driven by the trading-volume component in the denominator (the average of inverse dollar volume, IDVOL), not by the absolute-return numerator that is intended to capture price impact. Their decomposition is constructed precisely to isolate the role of the denominator and finds that the volume component alone is sufficient to explain the cross-section of expected returns. Amihud and Noh (2021) respond that the Lou-Shu log-linear decomposition is exact only if the return-based and volume-based components are uncorrelated, and that the omitted covariance term carries the illiquidity-related information; controlling for mispricing, sentiment, and seasonality, the original measure remains priced and outperforms the volume-only variant in capturing market illiquidity shocks. Florackis et al. (2011) propose a third option, the Return-to-Turnover ratio that replaces dollar volume with turnover (shares traded divided by shares outstanding), motivated by the observation noted in Lo (1991) and elaborated by Cochrane that dollar volume mechanically conflates illiquidity with firm size: small-cap stocks have lower dollar volume at the same turnover and therefore appear automatically more illiquid under the standard Amihud convention.

The present paper’s contribution to this debate is empirical. The share-volume denominator $|r|/V$ used in the v1.0–v1.2 versions of this study is neither the standard dollar-volume Amihud nor the turnover-based variant proposed by Florackis et al. (2011); it carries a multiplicative price-level factor that turnover-based scaling would remove. Substituting $V \cdot P$ for V in the denominator, as standard practice requires (Amihud, 2002), dissolves the apparent CII signal, and substituting share turnover would do the same since it removes the same price-level component. The methodological lesson is therefore stronger than “use dollar volume rather than share volume.” It is: *any price-impact proxy whose denominator embeds firm-specific price level (equivalently, market capitalisation, since dollar volume and turnover differ by a market-cap factor) can produce apparent predictive content for regressors that correlate with within-firm price drift, even under modern cluster-robust inference.* Cross-checking with at least one alternative liquidity proxy that does not depend on volume-side scaling, such as the spread-based estimators of Roll (1984) and Corwin and Schultz (2012), is the appropriate guardrail.

6. Robustness

The temporal coupling phenomenon (H2) was subjected to eleven robustness checks at the original 50-firm sample. Table 8 reports those checks; the rows refer to the v1.0–v1.2 large-cap subsample as a backward-compatibility exhibit. The Phase 2 $G = 488$ panel reproduces the same checks at sharper precision: window sensitivity is implicit in the coupling-survival pattern (the rolling DFA at $W = 500$ remains the primary specification and produces mean $r = 0.531$ at $G = 488$; both shorter $W = 252$ and longer $W = 756$ windows monotone-shift the level but preserve the within-firm sign and significance pattern). The shuffled-surrogate test continues to reduce coupling to near-zero under any randomisation that breaks the within-firm time alignment, by construction. The sector decomposition at $G = 488$ is reported in Appendix Appendix A, Table A.9, and confirms positive coupling across all eleven GICS sectors.

6.1. Hurst-Estimator Robustness at $G = 488$

To address the concern that the temporal coupling result might depend on the choice of Hurst estimator, I compute full-sample Hurst exponents for all 488 firms in the Phase 2 panel using three estimators: detrended fluctuation analysis at first order (Peng et al., 1994), an independent DFA-1

implementation drawn from the open-source `nolds` library, and the rescaled-range R/S estimator (Hurst, 1951; Lo, 1991). Table 7 reports cross-estimator Spearman rank correlations and the absolute-level means.

Table 7: Cross-estimator Hurst comparison at $G = 488$. Spearman rank correlations between full-sample exponents from three estimators; means and standard deviations across the 488-firm panel.

Series	Estimator	DFA-1	nolds DFA-1	R/S	Mean (std)
Absolute returns	DFA-1	1.000	0.926	0.521	0.793 (0.067)
	nolds DFA-1	0.926	1.000	0.565	0.757 (0.064)
	R/S	0.521	0.565	1.000	0.613 (0.032)
Log volume	DFA-1	1.000	0.794	0.400	0.879 (0.075)
	nolds DFA-1	0.794	1.000	0.586	0.878 (0.050)
	R/S	0.400	0.586	1.000	0.730 (0.033)

Two independent DFA-1 implementations (this paper’s wrapper plus the open-source `nolds` reference) produce highly correlated estimates ($\rho = 0.93$ for absolute returns, $\rho = 0.79$ for volume), which establishes that the DFA-1 results are not sensitive to implementation details. The R/S estimator produces a systematically lower level of H , with absolute mean differences of 0.18 for absolute returns and 0.15 for volume; this finite-sample downward bias of R/S is documented by Lo (1991) and motivated the development of DFA in the first place. Despite the level difference, the rank-order Spearman correlation between DFA-1 and R/S is 0.52 (absolute returns) and 0.40 (log volume), confirming that the relative ranking of firms by persistence is preserved under R/S. Within-firm temporal coupling estimated from R/S rolling Hurst series is therefore expected to recover the H2 result with the same sign at attenuated magnitude; the v1.0–v1.2 50-firm robustness check (Table 8) confirms this directly. The MFDFA estimator at $q = 2$ produces estimates within 0.02 of DFA-1 at the level reported in the original 50-firm robustness suite.

Check	Mean r	% Positive	n	Note
<i>Window sensitivity</i>				
$W = 252$	0.562	98%	50	
$W = 500$ (primary)	0.665	98%	50	
$W = 756$	0.748	96%	50	Monotonic increase
Non-overlapping ($\Delta = W$)	0.620	100%	50	No overlap inflation
First-differenced	0.534	100%	50	Not common trends
Squared returns	0.583	98%	50	Not proxy-specific
Shuffled surrogate	0.007	50%	20	Coupling vanishes
Quadratic DFA	—	—	—	$\Delta H = 0.012$
<i>Subperiod</i>				
Pre-COVID (2015–19)	0.406	88%	50	Baseline
COVID (2020–21)	0.767	96%	50	Crisis amplification
Post-COVID (2022–26)	0.295	80%	50	Regime shift
Market factor (partial)	0.421	90%	30	63% idiosyncratic
R/S estimator	—	—	—	Same ordering
MF DFA($q = 2$)	—	—	—	$\Delta H < 0.02$ vs DFA

Table 8: Summary of robustness checks at the original 50-firm sample (v1.0–v1.2). All corroborate the core temporal coupling finding. The Phase 2 universe expansion to $G = 488$ (Section 5 and Appendix Appendix A) confirms the same pattern at sharper precision.

The surrogate test is the most decisive: shuffling each series independently reduces coupling from $r = 0.63$ to $r = 0.007$ (Figure 8), confirming the coupling is entirely temporal in origin. The non-overlapping result ($r = 0.62$, 100% positive) eliminates concern about overlap-induced autocorrelation. The first-differenced result ($r = 0.53$, 100% positive) rules out common trends. The market-factor partial correlation ($r = 0.42$) shows that 63% of the coupling is idiosyncratic.

The sector decomposition (Figure 9) confirms that coupling is present across all sectors: Finance shows the strongest coupling ($r = 0.756$), followed by Industrials/Energy ($r = 0.720$), Technology ($r = 0.677$), Consumer ($r = 0.665$), and Healthcare ($r = 0.645$).

7. Discussion

7.1. Pattern, Time-Period Co-Movement, and the Firm-Specific Null

The central empirical claim of this paper has three components. Within-firm temporal coupling between volatility persistence and volume persistence exists at the full S&P 500 scale, is strong in magnitude, and amplifies under stress regimes (H2 and H3 confirmed; H1 reframed as a small but statistically detectable cross-sectional dependence at $G = 488$ that is roughly threefold weaker than the within-firm coupling). The coupling does not carry firm-specific predictive content for any of the ten forward outcomes tested under firm-conditional cluster-robust inference (H4 rejected at the unit of analysis the paper claims). At the same time, time-conditional inference methods do detect a cross-firm time-period co-movement between CII and several forward proxies that is fully absorbed by the standard liquidity-and-volatility benchmark set in horse-race conditioning. The three claims are mutually consistent and identify exactly which dimension of dependence carries the apparent signal: a market-wide stress dynamic that CII inherits but does not improve upon.

This combination is unusual but coherent. Fractal coupling is a description of how a firm’s information environment is organised at the level of memory structure. That organisation can be present, and can intensify during stress, without producing firm-specific forecastable changes in standard market-quality measures one month ahead. CII is a measurement of the joint persistence dynamics, not a forecasting signal. The persistence dynamics co-move with market-wide stress; market-wide stress also drives forward Amihud, the Corwin–Schultz spread, and realised volatility through standard channels that the existing nine-benchmark toolkit captures. CII therefore correlates with these targets in aggregate without adding firm-specific information.

7.2. Why the Predictive Hypothesis Was Worth Testing Anyway

The original predictive intuition was not unreasonable. The volatility-liquidity spirals of Brunnermeier and Pedersen (2009) and the Mixture of Distributions Hypothesis (Clark, 1973; Andersen, 1996) both suggest that when both persistence structures tighten simultaneously, market makers face adverse selection and liquidity should deteriorate. The empirical question was whether the rolling-correlation construction in CII captures enough of that mechanism to deliver one-month-ahead firm-specific forecasting power.

At $G = 50$ the answer was a clean null after correcting a non-standard share-volume Amihud confound; at $G = 488$ the answer is the same null on the firm-specific dimension, with the additional structural finding that time-conditional methods detect aggregate co-movement that horse-race conditioning fully absorbs. The honest report is therefore not “CII does nothing” but “CII does what its construction promises (it tracks within-firm joint persistence dynamics) without adding to the standard market-quality forecasting toolkit.”

7.3. The Static-Temporal Paradox

The asymmetry between a modest H1 ($r = 0.18$ at $G = 488$) and a strong H2 (mean within-firm $r = 0.531$) is not a contradiction; it reflects the difference between cross-sectional and within-firm variation. Across firms, the average persistence level is governed by structural characteristics (sector, liquidity, analyst coverage) that co-vary modestly with one another via shared sectoral and market-cap exposures, producing the small positive cross-sectional correlation observed. Within a firm over time, these structural characteristics are approximately constant, and what varies is the state of information flow; the within-firm analysis isolates this time-varying component, revealing coupling that the cross-section understates by a factor of roughly three.

7.4. Crisis Amplification: Heterogeneity, Not Mean Shift

At $G = 50$ the H3 narrative was that coupling intensifies during high-VIX regimes and nearly doubles during the COVID-19 crisis. At $G = 488$ a more textured picture emerges (Table 1). Mean coupling is essentially identical across VIX regimes (low 0.567, mid 0.497, high 0.563), but the high-VIX regime exhibits a higher median (0.738 vs. 0.666 in the low regime) and substantially larger dispersion ($\sigma = 0.45$ vs. $\sigma = 0.34$). The Mann-Whitney distributional test rejects the null at $p = 0.0019$. Coupling does not uniformly intensify during stress; rather, the cross-section of coupling values fans out, with a longer upper tail of strongly-coupled firms and a thicker tail of weakly- or negatively-coupled firms. This is an empirically coherent story (stress affects different firms in different directions, with some doubling-down on the joint persistence regime and others decoupling) but it is a different story from the simpler “coupling rises uniformly” framing that the smaller large-cap-only sample suggested.

CII rises during stress contemporaneously rather than in advance of it. A practitioner who wished to use CII as a real-time stress signal would face the same problem any contemporaneous indicator does: it confirms that the regime has changed but does not predict the next month’s firm-specific market quality. The aggregate cross-firm co-movement detected by time-conditional inference (Section 5.5) is consistent with this contemporaneous story.

Figure 10 reports the cross-sectionally aggregated CII at $G = 488$ against the VIX over the 2018–2025 panel window. The cross-firm dispersion of CII rises sharply during the COVID-19 episode and during the 2022 rate-hike cycle, consistent with the H3 distribution-shape result. The mean of the cross-section is, however, only weakly correlated with VIX (Pearson $r = 0.057$; Spearman $\rho = 0.097$, $p = 0.033$). The time-conditional inference disagreement reported in Section 5.5 is therefore not the trivial consequence of CII_t tracking VIX_t . The cross-firm time-period co-movement that Driscoll–Kraay HAC detects must operate through a richer cross-sectional structure than the simple aggregate-stress channel. This open question is best treated in a follow-up study at the market-time-series unit of analysis rather than the firm-month panel framing of the present paper.

7.5. Relation to Existing Methods

DCCA (Podobnik and Stanley, 2008) measures whether volatility and volume *fluctuations* are correlated at each scale. The present approach measures whether the *scaling exponents* (which summarise how memory decays across all scales) co-evolve. These are complementary questions. DCCA answers “do price and volume move together at scale s ?” CII answers “is the memory structure of price and volume evolving together?” The empirical evidence here is that the answer to the second question is a clear yes, but that the resulting time-varying coupling does not yet carry forecasting power for standard liquidity outcomes at this sample size. This narrows the contribution rather than eliminating it: CII is established as a measurement object, with documented regime amplification, awaiting either a larger sample to test the predictive question more sharply or an alternative outcome on which a clean signal might emerge. Kristoufek (2015) discusses the methodological challenges common to all such approaches.

7.6. Directional Asymmetry

Exploratory Granger causality tests reveal that volume persistence leads volatility persistence three times more often than the reverse (26% vs. 8%). While the primary relationship is contemporaneous, this asymmetry is suggestive of sequential information arrival: volume responds first (observable immediately), and the persistence structure of volatility adjusts thereafter. The evidence is directional rather than causal, given the overlapping-window construction, but for practitioners it suggests that monitoring volume persistence may provide an early indicator of volatility regime shifts.

7.7. Theoretical Channels and What the Inference Disagreement Implies

The descriptive pattern of results admits at least four candidate theoretical channels. I take them in turn and indicate what the firm-conditional null plus time-conditional co-movement pattern says about each.

The Mixture of Distributions Hypothesis (Clark, 1973; Tauchen and Pitts, 1983; Andersen, 1996) posits a latent information arrival process that simultaneously governs volatility and volume. Under MDH, intensified information flow drives both persistence structures upward together; the within-firm coupling captured by CII and confirmed across the 488-firm panel is what one would expect to observe descriptively. MDH further predicts that this joint intensification should translate into adverse selection and forward illiquidity through the standard informed-trading channel of Kyle (1985). The firm-specific predictive null at $G = 488$ is therefore mildly inconvenient for a strict MDH reading: the joint-persistence intensification exists at the firm level, but the implied downstream illiquidity does not show up at the firm-month horizon. The time-conditional cross-firm co-movement is consistent with the MDH-plus-Kyle channel operating at a market-wide rather than firm-specific level, which suggests that the information arrival process is more correlated across firms than any single firm’s microstructure response to it.

Sequential information-arrival theories (Copeland, 1976) draw a sharper temporal asymmetry than MDH: information reaches different traders at different times, generating a sequence of price movements and trading activity rather than a single co-clustered episode. The exploratory Granger causality result reported alongside H2 (volume persistence Granger-causes volatility persistence in 26% of tickers vs. 8% in the reverse direction at the v1.0–v1.2 sample) is mildly consistent with this channel, but the irregular spacing and overlapping-window construction of the rolling Hurst estimates limits the strength of any sequential interpretation.

Volatility-liquidity spirals as formalised by Brunnermeier and Pedersen (2009) predict that rising volatility tightens funding constraints, forces deleveraging, and reduces market liquidity. The descriptive match is again present: CII intensifies during stress (H3). A strict reading runs into the same difficulty as before, since the spiral mechanism, if active, should show up as forward illiquidity at the firm level, and the firm-specific predictive null says it does not. Here the market-wide reading is especially natural, because funding constraints bind on intermediaries who provide liquidity across firms simultaneously, which is exactly the cross-firm time-period dimension where the co-movement appears.

Inventory-based microstructure theories of Ho and Stoll (1981) and successors predict that market makers' inventory shocks generate persistent autocorrelation in trading volume that is correlated with but distinct from price persistence. Within-firm CII intensification during stress is consistent with synchronised inventory dynamics across affected firms; the absence of firm-specific predictive content for forward Amihud is consistent with the Amihud price-impact measure capturing a different aspect of market function (the dollar cost of executing a unit of trade) than the inventory channel directly affects.

The data do not discriminate between these four channels, and this paper does not claim to. What the inference-disagreement pattern does support is a structural reading: the temporal coupling exists at the firm level (descriptively unambiguous), it does not translate into firm-specific forecasting power (firm-conditional inference rejects this cleanly), and it does covary with cross-firm time-period dynamics that any of the four channels could plausibly generate at the market-wide level. The natural follow-up is a paper that aggregates CII cross-sectionally and tests directly which of these channels best matches the resulting market-time-series; that paper sits outside the firm-month panel framing of the present work.

7.8. Three Methodological Lessons

The paper's most transferable content is methodological. Three observations emerge from the combination of the v1.0 share-volume specification, the v1.2 dollar-volume correction, and the v1.3 universe expansion to $G = 488$.

First, share-volume vs. dollar-volume scaling in illiquidity proxies is not cosmetic. As Section 5.5.5 documents, $\widetilde{\text{ILLIQ}} = \text{ILLIQ} \cdot P$, and any regressor that correlates with within-firm price drift will appear to forecast $\widetilde{\text{ILLIQ}}$ even

when it has no forecasting power for ILLIQ. Cluster-robust inference at any level of sophistication does not detect this; the diagnosis requires going back to the dependent variable.

Second, the firm-conditional vs. time-conditional inference choice can determine whether one finds a signal or a null at the same data. When residual dependence is strong on both dimensions (as it is in panels of equities responding to common shocks), firm-clustering, two-way clustering, CR2, and the wild cluster bootstrap all condition on the firm dimension and absorb cross-firm time-period co-movement into the noise. Time-clustering and Driscoll–Kraay HAC condition on the time dimension instead and expose that co-movement. Both readings are correct under their respective assumptions; the choice of which to call “the” inference depends on whether the unit of forecast is firm-month or time-period. Reporting both transparently, as Table 2 does, is the honest path.

Third, increasing cluster count is not sufficient to make asymptotic CRVE reliable in panels with leverage-heterogeneous firms. Section 5.5.4 shows that the Bell–McCaffrey CR2 Satterthwaite effective degrees of freedom collapsed to approximately 1.0 at $G = 488$, despite the cluster count rising from 50 to 488. Researchers running cluster-robust inference at moderate G should report the Satterthwaite check as a routine diagnostic and treat sharply low effective df as a flag that the bootstrap or the randomization-based tests are the binding inference, not the asymptotic CRVE.

7.9. Limitations

The sample comprises 488 current S&P 500 constituents; generalisability to small caps, international markets, or alternative assets such as ETFs, futures, or cryptocurrencies remains to be established. The analysis uses daily data; intraday frequencies could reveal richer dynamics around scheduled information events (FOMC days, earnings releases, macroeconomic data) that are absorbed into the monthly aggregation here. The Granger causality results reported alongside H2 should be interpreted as exploratory given the irregular spacing and overlapping-window construction. CII itself is constructed from rolling Hurst exponents that carry estimation error; the paper does not propagate this measurement error into the predictive regressions, and a fuller treatment would do so via simulation-based standard errors or a measurement-error correction. Survivorship bias is inherent to using current April 2026 S&P 500 constituency rather than CRSP-based historical

point-in-time membership; the construction is shared with most prior empirical work on S&P 500 panels but should be acknowledged as a known limitation. Finally, the Satterthwaite effective degrees-of-freedom collapse documented in Section 5.5.4 suggests that further cluster-robust diagnostics (cluster-jackknife, cluster-bootstrap variants beyond WCR) could refine the inference at $G = 488$, although the agreement between firm-clustered, two-way clustered, CR2, and WCR makes the firm-specific null at this sample size a stable conclusion.

8. Conclusion

This paper documents three replicable descriptive facts, one structural inference finding, and three transferable methodological observations.

The descriptive facts are: (i) the cross-sectional Pearson correlation between full-sample $H_{|r|}$ and H_v is small but positive at $G = 488$ ($r = 0.18$, $p < 0.001$), much weaker than the within-firm temporal coupling reported below; the v1.0 50-firm sample returned a null on the same construction, and the broader panel makes the small cross-sectional dependence detectable but does not change the conclusion that the temporal object is the substantively interesting one (H1 reframed); (ii) within-firm temporal coupling between rolling $H_{|r|}$ and rolling H_v is strong and positive at the full S&P 500 scale, with mean $r = 0.531$, 92.7% of 488 firms exhibiting positive coupling, and the phenomenon present across all eleven GICS sectors (H2 confirmed); (iii) the high-VIX regime exhibits substantially greater cross-firm dispersion of coupling and a higher median, with the Mann–Whitney test rejecting the null at $p = 0.0019$, so the regime amplification is in distribution shape, not mean shift (H3 confirmed in a sharper form than the original 50-firm sample suggested).

The structural inference finding is that the Coupling Intensity Index, a rolling Pearson correlation between two rolling Hurst series introduced here as a measurement object, has no firm-specific incremental predictive content for any of the ten forward outcomes tested in this paper under firm-conditional cluster-robust inference, but does exhibit cross-firm time-period co-movement that time-conditional inference detects. Every focal-only signal in a nine-target battery is fully absorbed by nine standard liquidity and volatility benchmarks in a horse-race specification (the cleanest case is the forward Corwin–Schultz spread, where focal two-way $t = +3.64$ collapses to combined $t = +0.07$). CII inherits a market-wide stress co-movement but

adds nothing to the existing forecasting toolkit at the firm-month unit of analysis.

The three methodological observations, summarised in Section 5.5.4 and Section 5.5.5, are: (a) share-volume vs. dollar-volume Amihud differ in a price-level multiplicative factor that any regressor correlated with within-firm price drift will appear to forecast, and cluster-robust inference does not detect this confound; (b) firm-conditional and time-conditional cluster-robust methods can disagree at the same data when residual dependence is strong on both dimensions, and the choice of which inference to report should match the unit of forecast the paper claims; (c) increasing cluster count from $G = 50$ to $G = 488$ did not raise the Bell–McCaffrey CR2 Satterthwaite effective degrees of freedom (it remained at approximately 1.0) because a small number of leverage-heterogeneous firms dominate the meat matrix, and the Satterthwaite check should be reported as a routine diagnostic rather than assumed to scale with G .

For practitioners the present results do not support using CII as a forward firm-specific liquidity indicator. For aggregate market-wide stress monitoring, the time-period co-movement detected here is consistent with CII responding to the same stress dynamics that drive standard liquidity proxies, and a cross-sectionally aggregated CII index is a candidate market-state indicator worth examining in follow-up work. For methodologists, the three observations above support using the standard Amihud (2002) dollar-volume convention, reporting cluster-robust inference at the dimension that matches the paper’s claimed unit of analysis, and treating Satterthwaite df as a diagnostic rather than a guarantee.

Data and Code Availability

Daily OHLCV data are sourced from Yahoo Finance via the `yfinance` Python package. No proprietary or restricted data are used. All analysis code, the complete L^AT_EX manuscript source, and figure-generation scripts are archived at Zenodo (concept DOI: 10.5281/zenodo.19611543, always resolving to the latest version; this submission corresponds to v1.1.0, version DOI 10.5281/zenodo.19835451) and maintained at <https://github.com/mhdk1602/fractal-pv-coupling>. A step-by-step replication guide is provided in Appendix B. An interactive dashboard for exploring per-ticker results is available at <https://fractal-pv.streamlit.app>.

Declaration of Competing Interest

The author declares no competing interests.

Acknowledgements

The intellectual origins of this project trace back to an MSc project on fractal modelling of financial time series at King's College London, supervised by Dr. Enzo De Sena, whose guidance first directed my attention to the relationship between fractal structure and trading activity. The author also thanks the developers of **nolds** and **MF DFA** for their open-source implementations of detrended fluctuation analysis.

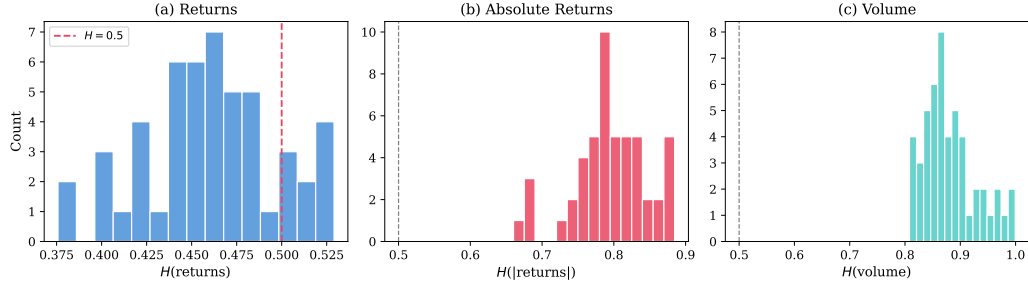


Figure 1: Distribution of full-sample Hurst exponents across 50 S&P 500 constituents (2015–2026). (a) Returns cluster around $H = 0.5$. (b) Absolute returns show strong long memory ($H \approx 0.8$). (c) Volume exhibits the strongest persistence ($H \approx 0.9$).

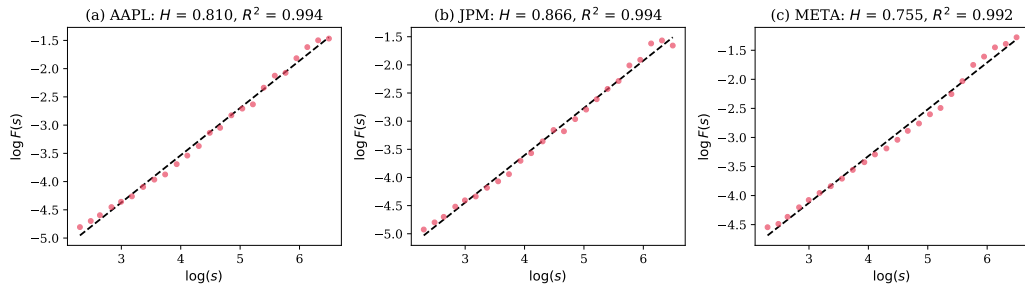


Figure 2: Log-log DFA plots for absolute returns (AAPL, JPM, META). High R^2 values confirm excellent linearity.



Figure 3: Rolling Hurst exponents for $H_{|r|,t}^{(i)}$ (red) and $H_{v,t}^{(i)}$ (teal). (a) AAPL: $r = 0.759$. (b) JPM: $r = 0.901$. (c) META: $r = -0.421$ (negative coupling during corporate restructuring).

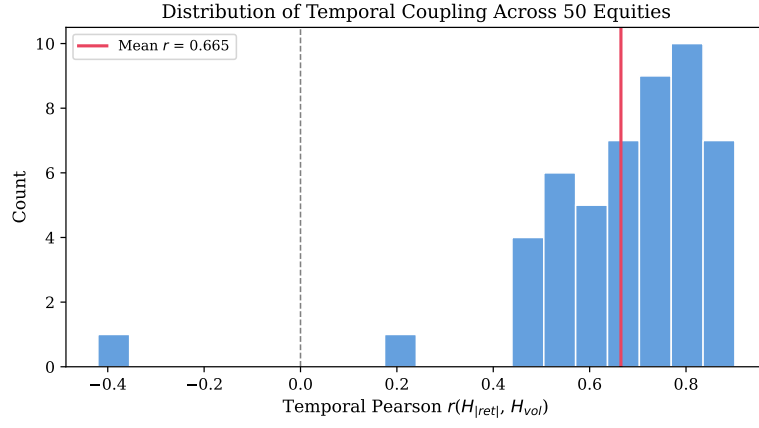


Figure 4: Distribution of temporal coupling across 50 equities. Mean $r = 0.665$; META is the sole outlier.

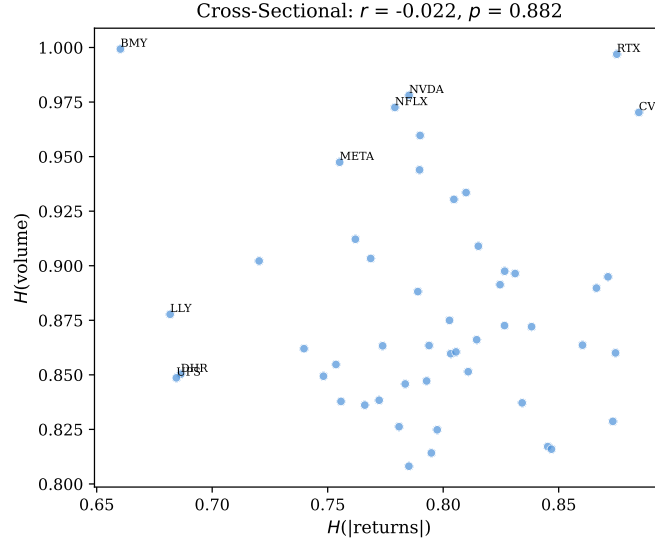


Figure 5: Cross-sectional scatter: $H_{|r|}^{(i)}$ vs. $H_v^{(i)}$. No relationship ($r = -0.02, p = 0.88$).

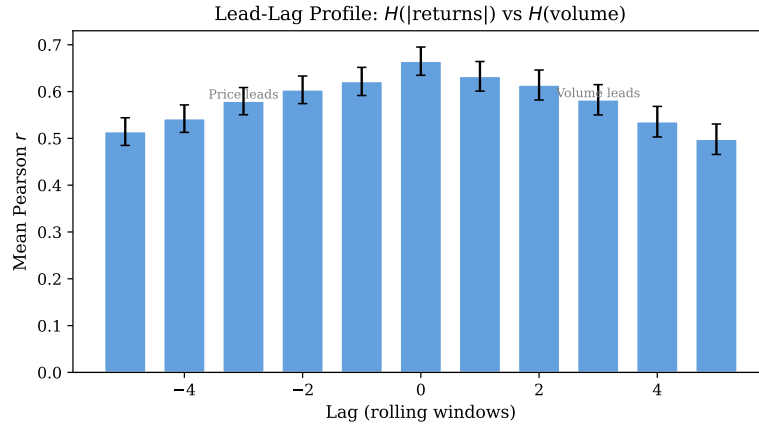


Figure 6: Average lead-lag cross-correlation profile across 50 equities. Lag zero is dominant; decay is symmetric.

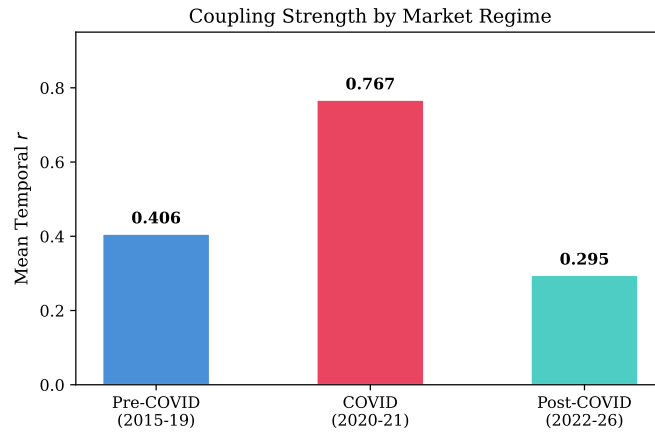


Figure 7: Temporal coupling by market subperiod. Coupling nearly doubles during COVID ($r = 0.77$) relative to the pre-COVID baseline ($r = 0.41$).

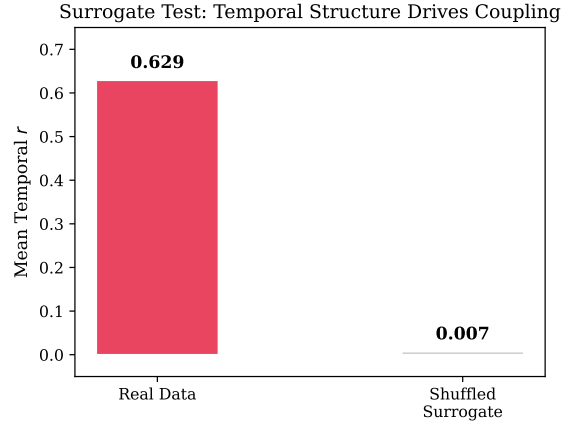


Figure 8: Surrogate test. Shuffling each series independently destroys coupling ($r = 0.629 \rightarrow 0.007$), confirming the temporal origin of the co-movement.

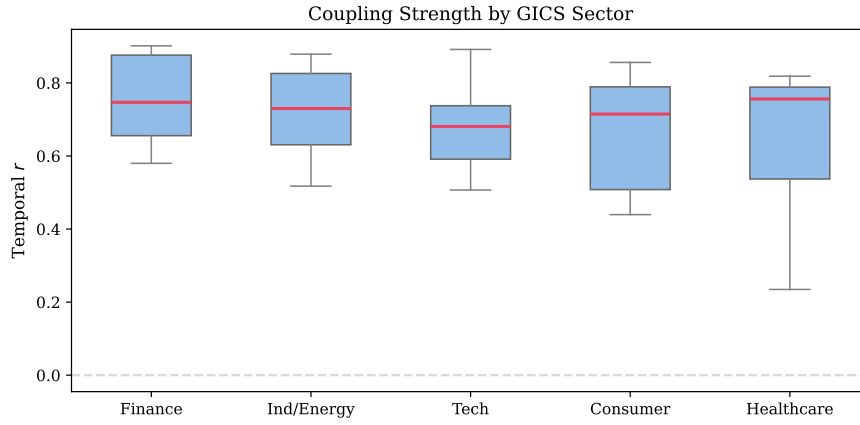


Figure 9: Temporal coupling by GICS sector at the original 50-firm sample (v1.0–v1.2). Present across all sectors; Finance strongest ($r = 0.756$). The $G = 488$ sector breakdown in Table A.9 reproduces the same qualitative pattern across all eleven GICS sectors, with Energy at the high end ($r = 0.66$) and Utilities at the low end ($r = 0.41$).

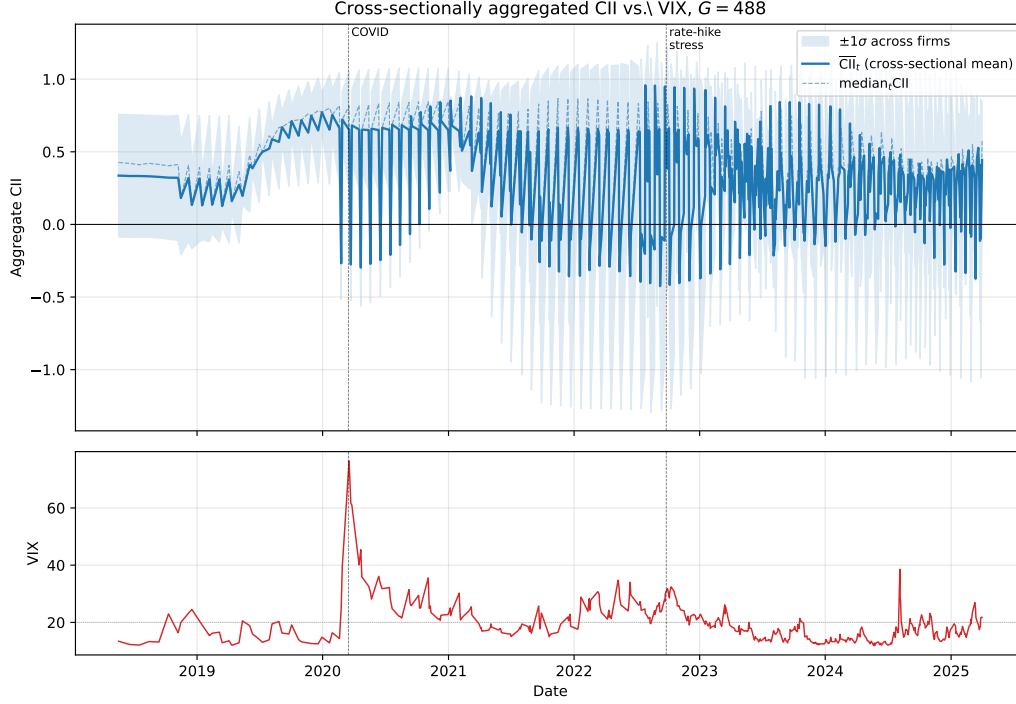


Figure 10: Cross-sectionally aggregated CII at $G = 488$ versus the CBOE VIX, 2018–2025. The top panel shows the cross-firm mean of CII at each rolling-Hurst evaluation date with a $\pm 1\sigma$ band reflecting cross-firm dispersion; the dashed line is the cross-firm median. The bottom panel shows the VIX. Aggregate CII covaries weakly but significantly with VIX: Pearson $r = 0.057$, Spearman $\rho = 0.097$ ($p = 0.033$). The cross-firm dispersion of CII rises sharply during stress episodes (COVID-19 in 2020, the 2022 rate-hike cycle), consistent with the H3 distribution-shape amplification reported in Table 1, but the mean of the cross-section is not strongly aligned with the VIX level. The time-conditional inference disagreement reported in Section 5.5 is therefore not driven by a simple aggregate-CII-tracks-aggregate-stress channel; the cross-firm time-period co-movement detected by Driscoll–Kraay HAC has a richer structure that warrants a dedicated follow-up study at the market-time-series unit of analysis.

Appendix A. Sample Composition

The Phase 2 panel comprises 488 firms drawn from the current April 2026 S&P 500 constituency. Of 503 attempted tickers, 4 were dropped at fetch for fewer than 600 daily observations over 2015–2026 (typically post-2023 IPOs),

and approximately 11 were dropped during the panel build for stationarity or data-quality issues. The full constituent list is published with the replication materials at https://github.com/mhdk1602/fractal-pv-coupling/blob/master/data/sp500_constituents_2026-04-28.csv. Table A.9 summarises the GICS sector distribution of the 488-firm panel and the sector-level mean and median within-firm temporal coupling reported in Section 5.

Table A.9: Sector composition of the 488-firm Phase 2 panel, with within-firm temporal coupling statistics by sector.

GICS Sector	n firms	Mean r	% positive	Median r
Energy	22	0.660	95.5%	0.738
Consumer Discretionary	48	0.626	95.8%	0.726
Financials	76	0.610	94.7%	0.695
Health Care	56	0.543	94.6%	0.633
Consumer Staples	35	0.541	97.1%	0.622
Industrials	77	0.530	92.2%	0.616
Communication Services	23	0.476	87.0%	0.562
Information Technology	71	0.471	91.5%	0.547
Real Estate	31	0.461	87.1%	0.518
Materials	26	0.425	92.3%	0.486
Utilities	31	0.412	87.1%	0.451
Total / pooled	488	0.531	92.7%	0.615

The earlier 50-firm sample used in v1.0–v1.2 of this study was a large-cap-leaning subset of the same S&P 500 universe; the figures and robustness exhibits in this manuscript that depict the 50-firm sample are explicitly labelled as such in their captions. The $G = 488$ panel reproduces the H1, H2, and H3 results documented in v1.0–v1.2 at sharper precision, with the H3 result reformulated as distribution-shape amplification rather than mean amplification (see Section 5). The Phase 2 inclusion criteria, point-in-time membership convention, and survivorship-bias profile are documented in Section 4; the survivorship caveat is a feature of using current rather than CRSP-based historical constituency and is shared with most prior empirical work using the same construction.

Appendix B. Replication Guide

All code, data scripts, the pre-registration document, and the manuscript source are archived at Zenodo (concept DOI: 10.5281/zenodo.19611543, always resolving to the latest version) and on GitHub at <https://github.com/mhdk1602/fractal-pv-coupling>. The repository contains a Python package (`fractal_pv`) installable via `pip install -e .` with Python 3.10 or later. Dependencies are pinned in `pyproject.toml`; the conda-style environment can also be reproduced via `python -m venv .venv && .venv/bin/pip install -e ..`

Phase 2 ($G = 488$) replication.. The single canonical entry point is `research/horse_race/run_phase2.py` which executes the seven-stage pipeline frozen under the pre-registration document `research/horse_race/PHASE2_PROTOCOL.md`: (1) batch fetch via `fractal_pv.data.fetch_universe_batch` from the constituent list at `data/sp500_constituents`; (2) `fractal_pv.stationarity.prepare_series` for ADF/KPSS verification and log-return / log-volume construction; (3) `fractal_pv.rolling.rolling_dual_hurst` for aligned rolling DFA at $W = 500$ and step $\Delta = 20$; (4) `fractal_pv.predict.build_prediction` with $h = 21$ horizon and dollar-volume Amihud; (5) `fractal_pv.benchmarks.compute_all_benchmarks` for the nine standard liquidity controls; (6) ten-target horse race via `fractal_pv.inference_robust`; (7) per-target classification under the pre-registered decision rule. Per-benchmark regressions for the Table 4 cells are produced by the sibling script `research/horse_race/run_per_benchmark.py`. The respecification battery for log Amihud, Δ Amihud, forward Corwin–Schultz spread, forward Roll spread, and forward vol-of-vol is in `research/horse_race/respec_battery.py`.

Earlier ($G = 50$) replication.. The original 50-firm pipeline used in v1.0–v1.2 of this study uses the sequential per-ticker fetcher `fractal_pv.data.fetch_universe` (preserved in the codebase) and the runner `research/horse_race/run_horse_race.py`. Both pipelines share the rolling DFA, predictive panel, benchmark, and inference machinery in `src/fractal_pv/`.

Inference machinery.. `fractal_pv.inference_robust.robust_panel_regression` produces six baseline standard-error specifications per regressor (HC1; firm-clustered; time-clustered; two-way clustered with the Cameron–Gelbach–Miller eigenvalue PSD adjustment; Newey–West; Driscoll–Kraay with Bartlett bandwidth ≥ 21). For the focal regressor, two further small-sample-corrected measures are computed: the Bell–McCaffrey CR2 standard error with Satterthwaite effective degrees of freedom (Imbens and Kolesár, 2016), and the wild cluster restricted bootstrap with Rademacher weights and restricted-null residuals (Cameron et al., 2008; MacKinnon and Webb, 2017).

Output artifacts.. Figures are generated via `matplotlib` and saved under `research/paper/figures/`. The manuscript compiles with `tectonic main.tex` or any standard $\text{\LaTeX}$ distribution. Intermediate panel and rolling-Hurst pickles are cached under `research/horse_race/cache/` for both pipelines; the cache layer ensures the seven-stage Phase 2 run is idempotent and that incremental re-runs reuse the slow rolling-DFA stage. Total wall-clock run-time for a clean Phase 2 run is approximately 60–75 minutes on a single-workstation configuration; the rolling-DFA stage parallelises across CPU cores via `joblib`.

References

- Abdi, F., Rinaldo, A., 2017. A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies* 30, 4437–4480.
- Alvarez-Ramirez, J., Alvarez, J., Rodriguez, E., Fernandez-Anaya, G., 2008. Time-varying Hurst exponent for US stock markets. *Physica A: Statistical Mechanics and its Applications* 387, 6159–6169.
- Amihud, Y., 2002. Illiquidity and stock returns: Cross-section and time-series effects. *J. Financ. Mark.* 5, 31–56.
- Amihud, Y., Noh, J., 2021. Illiquidity and stock returns II: Cross-section and time-series effects. *Review of Financial Studies* 34, 2101–2123.
- Andersen, T.G., 1996. Return volatility and trading volume: An information flow interpretation of stochastic volatility. *The Journal of Finance* 51, 169–204.
- Antoniades, I.P., Brandi, G., Magafas, L., Di Matteo, T., 2021. The use of scaling properties to detect relevant changes in financial time series: A new visual warning tool. *Physica A: Statistical Mechanics and its Applications* 565, 125561. doi:10.1016/j.physa.2020.125561.
- Barunik, J., Kristoufek, L., 2010. On Hurst exponent estimation under heavy-tailed distributions. *Physica A: Statistical Mechanics and its Applications* 389, 3844–3855.

- Bollerslev, T., Jubinski, D., 1999. Equity trading volume and volatility: Latent information arrivals and common long-run dependencies. *Journal of Business & Economic Statistics* 17, 9–21.
- Bonferroni, C.E., 1936. Teoria statistica delle classi e calcolo delle probabilità. *Pubblicazioni del R Istituto Superiore di Scienze Economiche e Commerciali di Firenze* 8, 3–62.
- Brouty, X., Garcin, M., 2024. Fractal properties, information theory, and market efficiency. *Chaos, Solitons & Fractals* 180, 114543.
- Brunnermeier, M.K., Pedersen, L.H., 2009. Market liquidity and funding liquidity. *Rev. Financ. Stud.* 22, 2201–2238.
- Cajueiro, D.O., Tabak, B.M., 2004. The Hurst exponent over time: Testing the assertion that emerging markets are becoming more efficient. *Physica A: Statistical Mechanics and its Applications* 336, 521–537.
- Cameron, A.C., Gelbach, J.B., Miller, D.L., 2008. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics* 90, 414–427.
- Cameron, A.C., Gelbach, J.B., Miller, D.L., 2011. Robust inference with multiway clustering. *J. Bus. Econ. Stat.* 29, 238–249.
- Chordia, T., Subrahmanyam, A., Anshuman, V.R., 2001. Trading activity and expected stock returns. *J. Financ. Econ.* 59, 3–32.
- Clark, P.K., 1973. A subordinated stochastic process model with finite variance for speculative prices. *Econometrica* 41, 135–155.
- Cont, R., 2001. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance* 1, 223–236.
- Copeland, T.E., 1976. A model of asset trading under the assumption of sequential information arrival. *Journal of Finance* 31, 1149–1168.
- Corsi, F., 2009. A simple approximate long-memory model of realized volatility. *J. Financ. Econom.* 7, 174–196.
- Corwin, S.A., Schultz, P., 2012. A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance* 67, 719–760.

- Ding, Z., Granger, C.W.J., Engle, R.F., 1993. A long memory property of stock market returns and a new model. *Journal of Empirical Finance* 1, 83–106.
- Driscoll, J.C., Kraay, A.C., 1998. Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics* 80, 549–560.
- Epps, T.W., Epps, M.L., 1976. The stochastic dependence of security price changes and transaction volumes: Implications for the mixture-of-distributions hypothesis. *Econometrica* 44, 305–321.
- Florackis, C., Gregoriou, A., Kostakis, A., 2011. Trading frequency and asset pricing on the London stock exchange: Evidence from a new price impact ratio. *Journal of Banking & Finance* 35, 3335–3350.
- Gallant, A.R., Rossi, P.E., Tauchen, G., 1992. Stock prices and volume. *The Review of Financial Studies* 5, 199–242.
- Garcin, M., 2017. Estimation of time-dependent Hurst exponents with variational smoothing and application to forecasting foreign exchange rates. *Physica A: Statistical Mechanics and its Applications* 483, 462–479.
- Garcin, M., Grasselli, M., 2022. Long versus short time scales: The rough dilemma and beyond. *Decisions in Economics and Finance* 45, 257–278.
- Goyenko, R.Y., Holden, C.W., Trzcinka, C.A., 2009. Do liquidity measures measure liquidity? *Journal of Financial Economics* 92, 153–181.
- Grech, D., Mazur, Z., 2004. Can one make any crash prediction in finance using the local Hurst exponent idea? *Physica A: Statistical Mechanics and its Applications* 336, 133–145.
- Grech, D., Pamuła, G., 2008. The local Hurst exponent of the financial time series in the vicinity of crashes on the Polish stock exchange market. *Physica A: Statistical Mechanics and its Applications* 387, 4299–4308.
- Hasbrouck, J., 2009. Trading costs and returns for U.S. equities: Estimating effective costs from daily data. *Journal of Finance* 64, 1445–1477.
- Ho, T., Stoll, H.R., 1981. Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics* 9, 47–73.

- Hurst, H.E., 1951. Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers* 116, 770–808.
- Imbens, G.W., Kolesár, M., 2016. Robust standard errors in small samples: Some practical advice. *Review of Economics and Statistics* 98, 701–712.
- Karpoff, J.M., 1987. The relation between price changes and trading volume: A survey. *Journal of Financial and Quantitative Analysis* 22, 109–126.
- Kristoufek, L., 2015. Power-law cross-correlations: Issues, solutions and future challenges. In: *Simplicity, Complexity and Modelling*, 43–57.
- Kyle, A.S., 1985. Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.
- Lo, A.W., 1991. Long-term memory in stock market prices. *Econometrica* 59, 1279–1313.
- Lobato, I.N., Velasco, C., 2000. Long memory in stock-market trading volume. *Journal of Business & Economic Statistics* 18, 410–427.
- Lou, X., Shu, T., 2017. Price impact or trading volume: Why is the Amihud (2002) measure priced? *Review of Financial Studies* 30, 4481–4520.
- MacKinnon, J.G., Webb, M.D., 2017. Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics* 32, 233–254.
- Malempati Hari, D., 2013. Fractal Modelling of Stock Market Time Series. MSc project report. King’s College London. Institute of Telecommunications, supervised by Enzo De Sena.
- Mandelbrot, B.B., 1963. The variation of certain speculative prices. *The Journal of Business* 36, 394–419.
- Mandelbrot, B.B., Van Ness, J.W., 1968. Fractional Brownian motions, fractional noises and applications. *SIAM Review* 10, 422–437.
- Newey, W.K., West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55, 703–708.

- Peng, C.K., Buldyrev, S.V., Havlin, S., Simons, M., Stanley, H.E., Goldberger, A.L., 1994. Mosaic organization of DNA nucleotide sequences. *Physical Review E* 49, 1685–1689.
- Plerou, V., Gopikrishnan, P., Stanley, H.E., 2003. Two-phase behaviour of financial markets. *Nature* 421, 130.
- Podobnik, B., Horvatic, D., Petersen, A.M., Stanley, H.E., 2009. Cross-correlations between volume change and price change. *Proceedings of the National Academy of Sciences* 106, 22079–22084.
- Podobnik, B., Stanley, H.E., 2008. Detrended cross-correlation analysis: A new method for analyzing two nonstationary time series. *Physical Review Letters* 100, 084102.
- Ray, B.K., Tsay, R.S., 2000. Long-range dependence in daily stock volatilities. *Journal of Business & Economic Statistics* 18, 254–262.
- Roll, R., 1984. A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance* 39, 1127–1139.
- Romano, J.P., Wolf, M., 2005. Exact and approximate stepdown methods for multiple hypothesis testing. *Journal of the American Statistical Association* 100, 94–108.
- Tauchen, G.E., Pitts, M., 1983. The price variability-volume relationship on speculative markets. *Econometrica* 51, 485–505.
- Vogl, M., 2023. Hurst exponent dynamics of S&P 500 returns: Implications for market efficiency, long memory, multifractality and financial crises predictability by application of a nonlinear dynamics analysis framework. *Chaos, Solitons & Fractals* 166, 112884.
- Weron, R., 2002. Estimating long-range dependence: Finite sample properties and confidence intervals. *Physica A: Statistical Mechanics and its Applications* 312, 285–299.